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COURSEWORK

IMPERIAL COLLEGE LONDON

MATH60141 INTRODUCTION TO GAME THEORY COURSEWORK 1

An Investigation of Goofspiel and Variations

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1 Introduction

Goofspiel, sometimes known as the *Game of Pure Strategy*, is a simple two-player, simultaneous, symmetric, and constant sum game which involves splitting a deck of cards into its 4 suits, *ace* (low, worth 1) to *king* (high, worth 13) and distributing them as follows:

- The *clubs* and *jokers* are set aside and not included in the game.
- The *hearts* are given to one player, and the *diamonds* to the other.
- Finally, the *spades* are shuffled and placed face-down in the centre of the players.

The premise of the game is that during each round, the card on the top of the spades pile is turned face-up, and both players must use their cards to '*bid*' on the shown central card. For example, if the central card is a king, the players may bid a king and a queen, in which case the player who bid the king would '*win*' the king, gaining its value (13 for the case of a king) in points. These three cards are then set aside for the rest of the game. This continues until all cards are out of play, when the player with the greatest score wins. If both players bid the same amount, neither player gains points.

This problem has been studied in detail – and while its analytical solution quickly becomes too complicated after $N = 5$, that is, a deck with 5 cards in each suit, numbered 1 through 5, it has been modelled extensively computationally, even going as far as $N = 16$.

Instead of viewing this problem through a traditional lens, we will first look at the game in its classical form (for N sufficiently small as to make non-computational analysis feasible) before exploring variations on the game, such as non-symmetric variations where each player starts with a different set of cards, and further investigating the dynamics of changing which cards each player has on their chances winning the game.

2 Formulation

2.1 Preliminaries

Suppose we are looking at the arbitrary $N = l$ case for *Goofspiel* – that is, there are l cards in each suit.

We first let $V = \{V_i : 1 \leq i \leq l\}$ be Player 1's cards with $V_i > 0$ (in the classical case $V = \{1, \dots, l\}$). We also, similarly, let $Y = \{Y_j : 1 \leq j \leq l\}$, and $P = \{P_k : 1 \leq k \leq l\}$ be Player 2's cards and the central cards respectively.

We can represent any single *Goofspiel* game or subgame uniquely using these sets – we define $G(V, Y, P)$ to be the *Goofspiel* game with the sets of cards V, Y and P .

Suppose in the game $G(V, Y, P)$, the card P_k is turned face-up. Player 1 and 2 now must decide a V_i and Y_j respectively to play, after which, Player 1 earns the following score:

$$s(V_i, Y_j, P_k) = \text{sgn}(V_i - Y_j)P_k, \quad (1)$$

where

$$\text{sgn}(x - y) = \begin{cases} 1 & x > y \\ 0 & x = y \\ -1 & x < y \end{cases} \quad (2)$$

Note that in reality, if Player 1 loses, they do not lose score, but instead Player 2 earns score. Due to the nature of the game as a constant-sum game, we can equivalently say that Player 1 loses score when making value calculations.

After these scores are given, the game is reduced into a subgame $G(V \setminus V_i, Y \setminus Y_j, P \setminus P_k)$.

Now, we can see that the payoff matrix X for Player 1 after the up-card P_k is drawn is defined recursively as follows:

$$X_{ij} = s(V_i, Y_j, P_k) + \mathcal{V}(\tilde{G}), \quad (3)$$

where $\tilde{G} = G(V \setminus V_i, Y \setminus Y_j, P \setminus P_k)$, and $\mathcal{V}(G)$ is the *value* of the game G .

Note that if $|V| = |Y| = |P| = 1$, then $\mathcal{V}(G(V, Y, P)) = s(V_1, Y_1, P_1)$ as these are the only cards left in each pile. We also note that $s(x, y, z) = -s(y, x, z)$ and $s(x, x, z) = 0$, that is, s is anti-symmetric in its first and second parameters.

We must also define $\mathcal{V}(G)$. We first introduce $G_k(V, Y, P)$ to be the game generated from $G(V, Y, P)$ after the up-card P_k is shown. Since there is an equal probability for each card to be drawn (though, this can be generalised to a case where each up-card has a set probability of being picked), we define:

$$\mathcal{V}(G(V, Y, P)) = \frac{1}{|P|} \sum_{k=1}^{|P|} \mathcal{V}(G_k(V, Y, P)), \quad (4)$$

that is, the average of all possible up-cards.

$\mathcal{V}(G_k(V, Y, P))$ is then the value of the game with payoffs X_{ij} , which we can solve for.

2.2 The semi-trivial case

To illustrate this model, we can look at the case $N = 2$, specifically, $V = Y = P = \{1, 2\}$, since the case $N = 1$ is trivial – the players do not get any choice in strategy and the game always ends in a draw.

$N = 2$ is no longer trivial as the player has to choose which card to bid first. However, their second bid is dependent on their first bid.

At the beginning the central card $p \in P = \{1, 2\}$ is turned up. Each value appears with probability $\frac{1}{2}$. If $p=1$: Let $V = Y = \{1, 2\}$ and indexing rows and columns by the bids 1, 2, we obtain the payoff matrix X by calculating each entry:

$$\begin{aligned} X_{11} &= s(1, 1, 1) + \mathcal{V}(G(\{2\}, \{2\}, \{2\})) = 0 + 0 = 0, \\ X_{12} &= s(1, 2, 1) + \mathcal{V}(G(\{2\}, \{1\}, \{2\})) = (-1) + 2 = 1, \\ X_{21} &= s(2, 1, 1) + \mathcal{V}(G(\{1\}, \{2\}, \{2\})) = 1 + (-2) = -1, \\ X_{22} &= s(2, 2, 1) + \mathcal{V}(G(\{1\}, \{1\}, \{2\})) = 0 + 0 = 0. \end{aligned}$$

Thus

$$X(P_1 = 1) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Similarly for $p = 2$ we have:

$$X(P_1 = 2) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Summing over we obtain $\mathcal{V}(G) = 0$, which means that although the $N = 2$ game involves strategic choice, its expected value remains zero.

3 Extensions

3.1 Asymmetric cases

In the previous part we assumed that the two players have same cards to play. We can consider the case where $V \neq Y$, but they have the same total card value. Suppose $N = 2$ but with $V = \{2, 3\}$ and $Y = \{1, 4\}$. At the beginning the central card $p \in P$ is turned up. Each value appears with probability $\frac{1}{2}$.

If $p = 1$: Let $V = \{2, 3\}$ and $Y = \{1, 4\}$ and index rows and columns by the bids 2, 3 and 1, 4 respectively. We obtain the payoff matrix X by again calculating each entry and obtain:

$$\begin{aligned} X(P_1 = 1) &= \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}, \\ X(P_1 = 2) &= \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}, \end{aligned}$$

$$\mathcal{V}(G(V, Y, P)) = (\mathcal{V}(G_1) + \mathcal{V}(G_2)) = \frac{1}{2}(-1 - 1) = -1$$

So after breaking the symmetry, one (Player 2) can shift the game value away from 0 (Even force a win in this case, as Player 2 can always bid 4 for $p = 2$ to get higher payoff.)

Now consider $N=3$ and Let

$$V = \{2, 3, 4\}, \quad Y = \{1, 3, 5\}, \quad P = \{1, 2, 3\},$$

with stage payoff $s(v, y, p) = p$ if $v > y$, $s(v, y, p) = -p$ if $v < y$, and $s(v, y, p) = 0$ if $v = y$. Using the same algorithm (but a lot more calculations!), we obtain $\mathcal{V}(G) = -\frac{1}{3}(\frac{14}{11} + \frac{10}{11} + \frac{14}{11}) \approx -1.15$, showing an advantage for Player 1. In this case, notice that if $P_1 = 2$, there is no strategy which Player 1 can play to force a win.

3.2 New scoring rule

Instead of being a winner-takes-all game, we can modify the payoff of Goofspiel to be proportionally dependent on the bids, and also make it zero-sum, i.e.

$$s(v, y, p) = p \frac{v - y}{v + y}.$$

If we apply this rule to the semi-trivial case, we see that the payoff matrices change to:

$$X(P_1 = 1) = \begin{pmatrix} 0 & \frac{1}{3} \\ -\frac{1}{3} & 0 \end{pmatrix}$$

$$X(P_1 = 2) = \begin{pmatrix} 0 & -\frac{1}{3} \\ \frac{1}{3} & 0 \end{pmatrix}$$

But the value is still 0. However, for the asymmetric case:

$$X(P_1 = 1) = \begin{pmatrix} \frac{1}{21} & \frac{2}{3} \\ -\frac{1}{6} & \frac{11}{21} \end{pmatrix}.$$

Here $\frac{1}{21}$ is a pure strategy equilibria hence $\mathcal{V}(G_1) = \frac{1}{21}$.

$$X(P_1 = 2) = \begin{pmatrix} \frac{11}{21} & -\frac{1}{6} \\ \frac{2}{3} & \frac{1}{21} \end{pmatrix},$$

Similarly $\mathcal{V}(G_2) = \frac{1}{21} \Rightarrow \mathcal{V}(G) = \frac{1}{2}(\mathcal{V}(G_1) + \mathcal{V}(G_2)) = \frac{1}{21}$ which means that instead of being forced to lose, player 2 has minor advantage under this rule!

4 Computational Approach

4.1 Analysis of individual games

Now, we look at approaching the problem from a computational standpoint in order to investigate our hypotheses for $N > 3$, as these are too complicated to analyse by hand. By employing a similar recursive method to the one used in Section 2 as well as introducing a few minor tweaks (such as storing sub-game values as to not require re-computation of values of smaller games) to save on computation time, we are able to analyse games up to $N = 6$ in a reasonable timespan.

For $N < 5$, the results aren't too interesting, being as expected, however for $N = 5$, our choice in strategy is a little less intuitive. While we would expect, usually, that betting a similar amount to the face-up card would be a wise choice, analysing the mixed strategy equilibriums of the $N = 5$ case shows a slightly different picture.

Table 1 compares the optimal mixed strategy for Player 1 in the game $G = G(V, Y, P)$ with $V = Y = P = [5]$, depending on the first face-up card. One can easily see that rather counter-intuitively, if a 2 is shown, we should *never* play a 2. Likewise for 3, we should *never* play a 3.

	Upcard				
	1	2	3	4	5
1	0.04699	0.18546	0.11817	0.12259	0.11230
2	0.83273	0	0.11880	0.07347	0.02410
3	0.12029	0.73749	0	0.19150	0
4	0	0.07705	0.76303	0.20434	0
5	0	0	0	0.40811	0.86359

Table 1: Optimal mixed strategies for $N = 5$

This is interesting, as one may assume that a good initial strategy is to bet the same as whatever card is shown. This clearly disproves that, and while we know that any fixed strategy that does not depend on the opponent's strategy can be beaten easily, it is still shocking that we should *never* bet the same as the upcard for $N = 2, 3$.

Table 2 compares the optimal mixed strategy for $N = 6$, and again we can see odd yet different behaviour. Similar to the $N = 5$ case, we should never bet $N - 1$ or $N - 2$ if the upcard is N , but this time we also should never play 1 if the upcard is N , a phenomenon that is also exhibited for all *even* $N > 5$. We also see that most of the time, if the upcard is 2, we should actually bet a 4, which may be unexpected.

	Upcard					
	1	2	3	4	5	6
1	0.16500	0	0.06547	0.09801	0.02726	0
2	0.57736	0.32534	0.13145	0.04582	0.08645	0.13833
3	0.25764	0.18141	0.16997	0.17340	0	0.00624
4	0	0.49326	0.29071	0	0.34613	0
5	0	0	0.34241	0.60814	0.02020	0
6	0	0	0	0.07462	0.51996	0.85542

Table 2: Optimal mixed strategies for $N = 6$

In actuality, we can see similar unexpected phenomena for $N > 6$ – we can see clear odd-even patterns for large upcards – one should never bet a card $< \frac{n}{2}$, where n is the value of the upcard, and of opposite parity to n ^[1].

4.2 Analysis of the meta-game

Now, we look further at the asymmetric game brought up in Section 3. Given a $k \in \mathbb{N}$ with $k \geq \frac{1}{2}N(N+1)$, $P = [N]$, $V = \{V_1, \dots, V_N\}$ to be picked and $Y = \{Y_1, \dots, Y_N\}$ such that $\sum_i V_i = \sum_j Y_j = k$, what is the optimal selection of V against any Y ?

We can tackle this problem by investigating the value of each possible V against each possible Y . Suppose $\tilde{V}_i$ and $\tilde{Y}_j$ are arbitrary choices of V and Y . Then, we define the payoff matrix X with $X_{ij} = \mathcal{V}(G(\tilde{V}_i, \tilde{Y}_j, P))$. Calculating the optimal strategy of this new meta-game defined by the payoff X will show us the optimal cards to pick.

By implementing this idea, we can calculate the optimal cards to pick for up to $N = 5$ in a reasonable timespan. Running our code, we find that the optimal strategy as implemented, at least for $N \leq 5$, is to pick $V = \{k - \frac{1}{2}N(N-1)\} \cup [N-1]$. That is, pick $N-1$ of the lowest cards, and then the highest card you can to reach k . For example, for $k = 21$, $N = 5$, this gives us $V = \{1, 2, 3, 4, 11\}$. This value is arrived at through setting up multiple games where the players have different bidding sets of cards, and finding the one which gives the first player the highest average payoff.

5 Conclusion

In investigating this change to a well understood game, we arrive at an interesting conclusion, which is that it is optimal to cluster the values of your cards towards the low end of the available values,

The choices we have made whilst producing this report and analysing this game do have some pitfalls, which could lead to subtly inaccurate results in our simulations and computations.

Our algorithm approaches optimal strategy finding based on maximising the "profit"

over the opponent; for example, if one turn of the game results in an 8-5 win for player one, the algorithm would determine that this result is superior to a 7-6 win, when in our understanding of the game, an overall win is a win and the eventual margin does not matter.

Our algorithm will generally produce the strategy set that gives player one a winning overall game, and maximising the payoff will, most of the time, produce this. However, it is possible, in some very specific cases, that attempting to maximise the overall score will be less optimal in terms of winning consistently than a strategy which focuses more on simply taking more tricks within the game.

Were we to have more time, we could attempt to optimise our algorithm to prioritise winning individual rounds and also maximising payoff, instead of attaching weight to beating the opponents by more. The new scoring rule implemented in 3.2, which is not present in our code, could help to prioritise winning the game over maximising our overall payoff, even though these two conditions are highly correlated.

Finally, we can investigate further properties of our modified game by taking our modified scoring system and raising it to a power α . This would allow us to investigate how the behaviour of our algorithm changes when payoffs are scaled arbitrarily, and whether this change leads to a different set of most optimal cards to select. Also, in this scenario, we could set $\alpha < 0$ and investigate how the behaviour of the game changes when the payoff for winning a specific turn in the game is inversely proportional to the value of the central card.

6 Acknowledgements + Appendix

All code used in the making of this project can be found online at <https://github.com/typtrue/gametheory-cw1>.

References

- [1] Glenn Rhoads and Laurent Bartholdi. Computer solution to the game of pure strategy. *Games*, 3:150–156, 11 2012. pages 7